$$\left[\frac{k(k+1)}{2}\right]^2 + (k+1)^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$$

$$\frac{k^2(k+1)^2}{4} + (k+1)^3$$

$$+ \frac{4(k+1)^3}{4}$$

$$(k+1)^2\left[\frac{k^2+4(k+1)}{4}\right]$$

$$\frac{k^2+4k+4}{4}$$

$$(k+1)^2\frac{(k+2)^2}{4} = \frac{(k+1)^2(k+2)^2}{4}$$

✓

we have proven that our base case, $n=1$ is true, we have also proven that our inductive step $k+1$ is true for any k. So we have proven that our formula is true for $n \geq 1$.

↓ if the formula holds at k